

Q1

(1-1)

1)

$$\begin{vmatrix} \lambda-4 & -2 \\ -2 & \lambda-1 \end{vmatrix} = \lambda^2 - 5\lambda = \lambda(\lambda-5) = 0$$

$$i) \lambda_1 = 0 \leadsto \begin{pmatrix} -4 & -2 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix} \leadsto u_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$ii) \lambda_2 = 5 \leadsto \begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix} \leadsto u_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

2)

$$A = PDP^T, \quad P = (u_1 u_2) \quad \& \quad D = \begin{pmatrix} 0 & 0 \\ 0 & 5 \end{pmatrix}$$

3)

$$\forall \lambda_i \geq 0 \Rightarrow A \text{ is PSD.}$$

(1-2)

1)

$$B^T B = \begin{pmatrix} 2 & 1 & 2 \\ 2 & -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 9 & 7 \\ 7 & 9 \end{pmatrix}, \quad B B^T = \begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 & 2 \\ 2 & -1 & 2 \end{pmatrix} = \begin{pmatrix} 8 & 0 & 8 \\ 0 & 2 & 0 \\ 8 & 0 & 8 \end{pmatrix}$$

2)

$$(\lambda-9)^2 = 49 \leadsto \lambda = 2, 16 \leadsto \sigma_1 = 4, \sigma_2 = \sqrt{2}$$

$$i) \lambda_1 = 16 \rightarrow \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \quad v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad ii) \lambda_2 = 2 \rightarrow \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \Rightarrow V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

3)

$$\begin{pmatrix} \lambda-8 & 0 & -8 \\ 0 & \lambda-2 & 0 \\ -8 & 0 & \lambda-8 \end{pmatrix} = (\lambda-2) \begin{pmatrix} \lambda-8 & -8 \\ -8 & \lambda-8 \end{pmatrix} = \lambda(\lambda-2)(\lambda-16)$$

$$i) \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad ii) \begin{pmatrix} 3 & 0 & 4 \\ 0 & 0 & 0 \\ 4 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad u_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$iii) \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad u_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \Rightarrow U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{pmatrix}$$

4)

$$\Sigma = \begin{pmatrix} 4 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{pmatrix} \leadsto B = U \Sigma V^T$$

(1-3)

1)

$$(\lambda - 5)^2 = 9 \quad \lambda = 2, 8$$

$$\text{i) } \lambda_1 = 2: \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \leadsto u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \text{ii) } \lambda_2 = 8: \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \leadsto u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

2)

$$\text{PC1: } u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \text{ PC2: } u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

3)

$$\text{PC1: } \frac{8}{10} = 80\% \quad , \quad \text{PC2: } \frac{2}{10} = 20\%$$

4)

20%

Q2

(2-1)

1)

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{4}, P(C) = \frac{1}{4}$$

2)

$$\mathcal{H}(X) = - \left\{ \frac{1}{2} \cdot (-1) + \frac{1}{4} \cdot (-2) + \frac{1}{4} \cdot (-2) \right\} = \frac{3}{2}$$

3)

$$\# \text{ of } A, B, C = 5, 5, 6 \leadsto \max \{ \mathcal{H}(X) \} = - \left\{ 2 \cdot \frac{5}{16} \cdot \log_2 \frac{5}{16} + \frac{6}{16} \log_2 \frac{6}{16} \right\} \approx 1.579$$

↳ 현재 데이터의 엔트로피는 1.5로 최댓값에 근접한 불확실성이 높은 상태.

(2-2)

1)

$$D_{KL}(P \parallel Q) = 0.5 \log_2 2 + 0.3 \log_2 0.6 + 0.2 \log_2 0.8 \approx 0.215$$

$$D_{KL}(Q \parallel P) = 0.25 \log_2 0.5 + 0.5 \log_2 \frac{2}{3} + 0.25 \log_2 1.25 \approx 0.199$$

∴ $D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P)$ 이므로 2개의 정의를 만족하지 않음.

2)

$D_{KL}(P \parallel Q) \rightarrow$ 확률분포 P 를 Q 로 근사하여 발생하는 정보량의 손실 ; 두 분포가 20이

$$\hookrightarrow P(X) = Q(X) \Leftrightarrow D_{KL}(P \parallel Q) = 0$$

Q3.

(3-1)

1)

$$\bar{x} = 3, \bar{y} = 5 \leadsto S_{xx} = 4 + 1 + 0 + 1 + 4 = 10, S_{xy} = 6 + 2 + 0 + 1 + 8 = 17$$

2)

$$\hat{\beta}_1 = \frac{17}{10}, \hat{\beta}_0 = 5 - \frac{17}{10} \cdot 3 = -\frac{1}{10} \Rightarrow \hat{y} = -\frac{1}{10} + \frac{17}{10}x$$

3)

$$\hat{y} = \{1.6, 3.3, 5, 6.7, 8.4\} \Rightarrow e = \{0.4, -0.3, 0, -0.7, 0.6\}$$

$$\hookrightarrow SSE = 0.16 + 0.09 + 0 + 0.49 + 0.36 = 1.1 \Rightarrow R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} \approx 0.9633$$
$$SST = 9 + 4 + 0 + 1 + 16 = 30$$

4)

R^2 값이 1에 가까우므로 맞추울 설명.

(3-2)

1)

$$X_1\text{'s VIF} = \frac{1}{1 - 0.9} = 10$$

2)

$VIF \geq 10 \Rightarrow X_1$ 의 제거를 강력히 고려

Q4

(4-1)

1)

$$\begin{cases} d(f, A_1) = \sqrt{18}, d(f, A_2) = 0.5 \\ d(f, B_1) = \sqrt{2}, d(f, B_2) = \sqrt{2}, d(f, B_3) = 2 \end{cases}$$

2)

$$3\text{-nn} \rightarrow A_2, B_1, B_2 \Rightarrow f \in B$$

3)

$$\sum w_A = 2, \sum w_B = \sqrt{2} \Rightarrow f \in A$$

4)

결과가 다름 $\rightarrow$ 오히려 KNN 과 다르게 weighted KNN은 거리 분포의 균질도 고려하기 때문

Q5

(S-1)

1)

$$\begin{cases} d(C_1, P_i) \rightarrow 0, 2, \sqrt{2}, \sqrt{61}, \sqrt{24}, 6\sqrt{2} \leadsto C_1 \rightarrow P_1, P_2, P_3 \\ d(C_2, P_i) \rightarrow \sqrt{61}, \sqrt{85}, \sqrt{61}, 0, 1, 1 \leadsto C_2 \rightarrow P_4, P_5, P_6 \end{cases}$$

2)

$$C_1^{new} = (\frac{17}{3}, 2), \quad C_2^{new} = (\frac{25}{3}, \frac{25}{3})$$

3)

반응 $\times \rightarrow$ 알고리즘 수정.

(S-2)

1)

$$d(P_1, P_2) = 1, d(P_1, P_3) = 1, d(P_1, P_4) = \sqrt{2} \leadsto \{P_1, P_2, P_3, P_4\}$$

$$d(P_5, P_6) = 1, d(P_6, P_8) = \sqrt{2} \rightarrow \{P_5, P_6, P_8\}$$

$$d(P_7, P_8) = 1 \rightarrow \{P_7, P_8\}$$

2)

$$\begin{cases} \text{core} : P_1, P_2, P_3, P_4, P_5, P_6, P_8 \\ \text{border} : P_7 \\ \text{noise} : P_9 \end{cases}$$

3)

$$\{P_1, P_2, P_3, P_4\}, \{P_5, P_6, P_7, P_8\}$$

$\rightarrow$ DBSCAN 은 $\epsilon, \text{min-pts}$ 값만 설정해주면 클러스터를 자동으로